COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	16/07/26	**DATE SUBMITTED:**	16/07/26
NAME:	Mike Justin A. Cabangon	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. HOA 2.1 Arrays, Pointers, and Dynamic Memory Allocation

1. PROCEDURE OUTPUT

What procedures did you perform?

INPUT

```cpp
#include <iostream>
#include <string.h>

class Student{
    private:
        std::string studentName;
        int studentAge;
    public:

//Constructor
    Student(std::string newName = "John Doe", int newAge = 18) {
        studentName = newName;
        studentAge = newAge;

        std::cout << "Constructor was called" << std::endl;
}

//Destructor
    ~Student(){
        std::cout <<"Destructor was called" <<std::endl;
}

//Copy Constructor
    Student (const Student &copyStudent){
        std::cout << "Copy Constructor Called" << std::endl;
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
}

```

```
30  //Display Attributes
31      void printDetails(){
32          std::cout << this->studentName << " " << this->studentAge << std::endl;
33  }
34
35  //Copy Assignment Operator
36
37
38  };
39
40  int main() {
41      Student student1 ("Mike", 19);
42      Student student2 (student1);
43      Student student3;
44      student3 = student2;
45
46  return 0;
47  }
48
```

OUTPUT

```
Constructor was called
Copy Constructor Called
Constructor was called
Destructor was called
Destructor was called
Destructor was called
```

OBSERVATION

The Student class stores a student's name and age.
The constructor sets the values(“John Doe”, 18), the copy constructor copies another object's data, and the destructor runs when an object is deleted. In main(), one object is created, another is copied from it, and a third object is assigned the copied values.

Table 2-1: Initial Driver Program

INPUT

```cpp
#include <iostream>
#include <string.h>

class Student{
    private:
        std::string studentName;
        int studentAge;
    public:

//Constructor
    Student(std::string newName = "John Doe", int newAge = 18) {
        studentName = newName;
        studentAge = newAge;

        std::cout << "Constructor was called" << std::endl;
}

//Destructor
    ~Student(){
        std::cout <<"Destructor was called" <<std::endl;
}

//Copy Constructor
    Student (const Student &copyStudent){
        std::cout << "Copy Constructor Called" << std::endl;
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
}

```

```
30  //Display Attributes
31      void printDetails(){
32          std::cout << this->studentName << " " << this->studentAge << std::endl;
33  }
34
35  //Copy Assignment Operator
36
37
38  };
39
40  int main() {
41      const size_t j = 5;
42
43      Student studentList[j] = {};
44      std::string namesList[j] = {"Mike", "Callum", "Ethan", "Sophie", "Ashley"};
45      int ageList[j] = {19, 20, 21, 23, 25,};
46
47  return 0;
48  }
49
```

OUTPUT

```
Constructor was called
Constructor was called
Constructor was called
Constructor was called
Constructor was called
Destructor was called
Destructor was called
Destructor was called
Destructor was called
Destructor was called
```

OBSERVATION

The code first started to execute in main()
-In line 41 the compiler initialized an array with a size of 5.
-In index 0, “John Doe” was assigned to studentName and 18 was assigned to studentAge.
-line 15 prints out “Constructor was called” implying that a Constructor was executed.
-

Table 2-2: Modified Driver Program With Student Lists

INPUT

```cpp
#include <iostream>
#include <string.h>

class Student{
    private:
        std::string studentName;
        int studentAge;
    public:

//Constructor
    Student(std::string newName = "John Doe", int newAge = 18) {
        studentName = newName;
        studentAge = newAge;

        std::cout << "Constructor was called" << std::endl;
}

//Destructor
    ~Student(){
        std::cout <<"Destructor was called" <<std::endl;
}

//Copy Constructor
    Student (const Student &copyStudent){
        std::cout << "Copy Constructor Called" << std::endl;
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
}

```

```cpp
//Display Attributes
    void printDetails(){
        std::cout << this->studentName << " " << this->studentAge << std::endl;
}

//Copy Assignment Operator


};

int main() {
    const size_t j = 5;

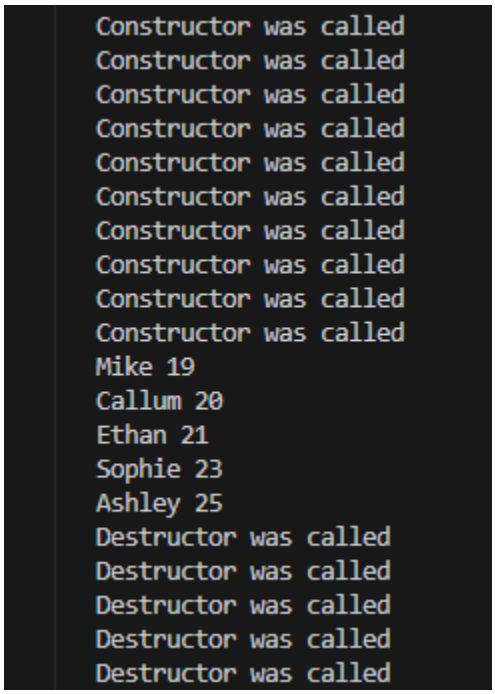
    Student studentList[j] = {};
    std::string namesList[j] = {"Mike", "Callum", "Ethan", "Sophie", "Ashley"};
    int ageList[j] = {19, 20, 21, 23, 25,};

    for (int i = 0; i < j; i++) { //loop A
        Student *ptr = new Student (namesList [i], ageList [i]);
        studentList [i] = *ptr;
    }

    for (int i = 0; i < j; i++) { //loop B
        studentList [i] .printDetails() ;

    }

    return 0;
}
```

OUTPUT


```
Constructor was called
Constructor was called
Constructor was called
Constructor was called
Constructor was called
Constructor was called
Constructor was called
Constructor was called
Constructor was called
Constructor was called
Mike 19
Callum 20
Ethan 21
Sophie 23
Ashley 25
Destructor was called
Destructor was called
Destructor was called
Destructor was called
Destructor was called
```

OBSERVATION

Table 2-3:Final Driver Program

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

COMPUTER ENGINEERING

QUEZON CITY

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.